

Radix & bit width

Hardware does not store “thirteen.” It stores a fixed-width bit pattern that you may read as binary

Same bits, many readings

- A width is a hard budget: eight bits is not the same box as four bits
- Inside one width, the pattern is shared
- Binary, hex, and unsigned decimal are different spellings of those bits
- Signed two's complement is another reading of the same pattern
- Change the width, and the meaning can change even if you type the same digits

Browser lab

Digital Design and Verification Platform

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INTERACTIVE TOOL

Radix / bit-width converter

Convert among binary, hex, unsigned, and two's-complement decimal for a chosen width — backed by the HDL engine Value type and `parseLiteral`. Watch overflow and bit patterns update together.

Starter example: 8-bit value 42 / 0x2A. Change width, type hex or signed decimal, or click bits.

Load starter example

Challenges0 / 22 cleared

○ Binary → unsigned

Intro · 8-bit · Enter binary

0000_1101. What is the unsigned decimal?

○ Hex FF unsigned

Intro · 8-bit · Set width 8 and hex FF.

Read the unsigned decimal.

○ Hex FF signed

Intro · 8-bit · With width 8 and pattern 0xFF, what is two's-complement signed?

○ Encode -5

Core · 8-bit · Using signed decimal, enter -5 at width 8. Confirm binary ends with ...1011.

○ Unsigned overflow

Core · 4-bit · Width 4: try unsigned decimal 20. Truncation should show pattern 0100 (=4).

○ Nibble 0xA

Core · 4-bit · Width 4, set hex A.

Signed value should be -6.

○ 16-bit -1

Core · 16-bit · Width 16: enter signed -1. Hex should be FFFF.

○ MSB is sign

HDL · 8-bit · Width 8: set only the MSB (bit 7) to 1. Signed should be -128.

○ Signed wrap

○ Hex 0x2A

○ Binary 10101010

○ Decimal 42

Binary → unsigned: Enter binary 0000_1101. What is the unsigned decimal?

Start selected

Show hint

Next

Stop

Reset progress

Idle

Radix converter starter

learn_digital — radix converter

Workbook practice

- In the workbook track, pick one concrete conversion and do it by hand
- For example, take unsigned decimal thirteen at width eight
- Write the eight-bit binary, then the hex
- Then flip to signed thinking: what signed value is the all-ones pattern at that width?
- Write one pitfall you will watch for in later RTL

Pitfalls to watch

- Do not treat hex digits as if the width were infinite
- Do not confuse “the number I typed” with “the bits the hardware keeps.” And remember
- When you write RTL later

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, finish a few challenges after the starter
- On paper, finish at least one worked conversion and name one pitfall
- When you are ready, take the short quiz, then continue to two's complement